

GoalForge

Current State Review Report

Date:	2026-03-14
Reviewer roles:	Lead Product Manager + Senior Software Engineer
Prior review:	2026-03-13 (PR #24)
Scope:	Backend (apps/api) Frontend (apps/web) CI setup

Changes Since Previous Review (2026-03-13)

The following areas have changed since the prior agent session. New findings are marked [NEW].

- Auth exception handling partially hardened — `_decode_token` in `auth.py` now catches `jwt.ExpiredSignatureError` and `jwt.InvalidTokenError` specifically, in addition to a remaining broad fallback.
- Dependency updates — React 19, Vite 7, TypeScript 5.9, Tailwind CSS v4, axios 1.13.6 (`apps/web/package.json`).
- No new frontend tests added — the testing gap from the prior review remains.
- `Dashboard.tsx` remains monolithic — 951 lines with no decomposition.
- No pagination added — goals listing API still returns all records without limits.

1) Lead Product Manager Review

Product Strengths

Clear core value proposition

GoalForge
(README.

Coherent motivational system

Star point:
Analytics.t

Foundational user journey exists

Sign-in/sig

[NEW] Active dependency maintenance

Frontend c
with low d

Product Risks / Gaps

High feature concentration in one UI file

Dashboard
roadmap c

No frontend automated tests

Backend t
UX change

Scalability gap in goal listing API

Goals API
users.

Operational reliability mismatch between local and CI defaults

Local bac
(.github/w

[NEW] Silent sprint generation failure on milestone advance

complete_
AIGenerat
instead of

PM Priority Recommendations (without removing shipped features)

- [P0] Fix wrong exception class in complete_milestone (ValueError -> AIGenerationError).
- [P0] Add frontend test baseline for critical flows (goal creation, task completion, status update).
- [P0] Split Dashboard into composable feature components.
- [P1] Add paginated goals listing and front-end support.
- [P1] Add explicit failure UX for profile/AI-related fetch degradation.
- [P1] Migrate FastAPI startup handler from deprecated on_event to lifespan pattern.
- [P2] Define product analytics events (activation, weekly retention, milestone completion rate).

2) Senior Software Engineer Review

Engineering Strengths

Clean backend layering and typed contracts

Good separ

[NEW] Improved auth exception specificity

_decode_t
differentia

Reasonable security foundations

JWT verifi

from env.

AI output schema constraints present

Gemini re
malformed

Backend test suite is healthy

23 integra

[NEW] CI pipeline is comprehensive

CI runs ba

Engineering Issues and Improvement Opportunities

[P0 BUG — NEW] complete_milestone catches the wrong exception class

File: apps/api/main.py | **Line:** 657

Current: except ValueError as exc:

Correct: except AIGenerationError as exc:

Impact: sync sprint generation failures propagate as unhandled 500 errors instead of the intended actionable 502 responses.

auth.py retains broad fallback exception in token decode path

Specific h
except Ex

[NEW] Deprecated FastAPI startup event handler

@app.on_
in test ou
constructo

Rate-limit keying strategy can cause noisy-neighbor effects

_user_key
path varia

Frontend reliability and maintainability pressure

Monolithi
Dashboard

API scalability and resilience

No pagina
durable re

Test and quality asymmetry

Backend h

Recommended Technical Roadmap

- [P0] Fix complete_milestone: change except ValueError to except AIGenerationError.

- [P0] Remove/scope remaining broad except Exception in `_decode_token`; add structured logging.
- [P0] Add frontend tests for highest-risk interactions (goal creation, task completion, milestone advance).
- [P1] Migrate `@app.on_event('startup')` to FastAPI lifespan pattern.
- [P1] Refactor `Dashboard.tsx` into domain-focused components/hooks.
- [P1] Introduce shared data-fetch hooks to remove duplication between Dashboard and Analytics.
- [P2] Add cursor/offset pagination for `/users/{user_id}/goals`.
- [P2] Add observability metrics for AI generation latency/failure rate.

3) Current Validation Snapshot (2026-03-14)

Frontend Baseline (apps/web)

<code>npm ci</code>	✓ dependencies installed cleanly
<code>npm run lint</code>	✓ pass — 0 warnings
<code>npx tsc --noEmit</code>	✓ pass — no type errors
<code>npm run build</code>	✓ pass — 323 KB JS bundle, gzip 97 KB

Backend Baseline (apps/api)

`RATE_LIMIT_ENABLED=false GEMINI_API_KEY=dummy DATABASE_URL=sqlite+aiosqlite:///memory:`

`python -m`

✓ 23/23 tests passed — 2 deprecation warnings (`on_event` startup handler)

4) Executive Summary

GoalForge is in strong MVP-to-beta shape: coherent product loop, good backend structure, and a comprehensive CI pipeline. Since the previous review (2026-03-13), the most notable improvement is partial auth exception specificity in `_decode_token`.

One new P0 bug was discovered: `complete_milestone` catches `ValueError` instead of `AIGenerationError`, meaning synchronous sprint generation failures surface as unhandled 500 errors rather than the intended actionable 502 responses.

Highest-Leverage Next Work

- [P0 — immediate] Fix wrong exception class in `complete_milestone` (1-line change).
- Frontend quality uplift: add tests + decompose `Dashboard.tsx`.
- Security/observability hardening: remove broad auth fallback, migrate startup handler.
- Scalability guardrails: add pagination + improve async task reliability.

All improvements can be delivered incrementally without removing existing shipped behavior.
The codebase is well-structured and ready for the next phase of quality hardening.